DebtMap

Local AI Debt Tracker & Payoff Planner

DAY 09 | JULY 6, 2026

Map your debt. Plan the escape. Local AI guides the way.

Category	Details
Startup Name	DebtMap
Type	Offline AI Debt Tracker & Payoff Planner (Mobile-Only)
Stack	Vanilla HTML/JS + Ollama
Models	llama3.2, mistral, gemma2, qwen2.5
Algorithms	Avalanche (highest APR first) + Snowball (lowest balance first)
Privacy	100% Local — Financial Data Never Leaves Device
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Series	Daily Local LLM Startup — Day 09
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WHITEPAPER

DebtMap: Privacy-First Debt Intelligence on Local AI

Abstract

DebtMap is a mobile-first, offline debt tracker and payoff planner powered by a locally-running LLM. It tracks multiple debts with APR, balance, and minimum payment data; computes Avalanche and Snowball payoff strategies; projects payoff dates and total interest; and generates personalised AI financial coaching — all without transmitting any financial data to external servers.

1. Problem Statement

Debt management apps (Tally, Undebt.it, Debt Payoff Planner) require account creation and either link to bank accounts or store debt data on cloud servers. For users managing sensitive financial situations — bankruptcy recovery, undisclosed debt, privacy-sensitive income sources — cloud storage of debt data is unacceptable. The financial data category has the highest sensitivity of any personal data type, making local-first architecture especially critical.

2. Solution

- Multi-debt tracking — name, type, balance, APR, minimum payment per debt
- Progress bars — paid-off percentage per debt visualised from original balance
- Avalanche strategy — sorts by highest APR first; minimises total interest paid
- Snowball strategy — sorts by lowest balance first; maximises psychological wins
- Extra payment allocation — user sets extra monthly budget; applied to top-priority debt
- Payoff date projection — closed-form amortisation formula per debt
- Total interest estimate — projected interest cost per debt under current payment plan
- AI financial coaching — personalised insight on biggest impact action + motivation
- Payment history — timestamped ledger of all recorded payments

3. Technical Architecture

Layer	Technology	Role
UI	Vanilla HTML5/CSS3 (mobile-only)	3-tab: Debts, Plan, History. Bottom-sheet modals. Max-width 480px
Logic	Vanilla JS (ES6+)	Debt CRUD, amortisation calc, strategy sort, payment recording
Storage	localStorage	Two arrays: debts (with live balance) + payments (ledger)
AI Runtime	Ollama local process	Financial coaching prompt: action + interest comparison + motivation
Amortisation	Closed-form log formula	months = -log(1-(B*r/P)) / log(1+r); handles r=0 edge case

4. Market Opportunity

Total US consumer debt exceeded USD 17.5 trillion in 2024. The debt management app market is growing at 12% CAGR, driven by rising credit card balances and student loan complexity. The underserved segment is privacy-sensitive users — estimated at 15-20% of the financially distressed population — who refuse cloud-linked apps but would use a local-only tool. DebtMap targets this gap with a zero-account, zero-sync, zero-cloud solution.

5. Monetization

Tier	Price	Offering
Free	USD 0	Unlimited debts, both strategies, basic projections
Pro	USD 5/mo	Unlimited AI insights, PDF payoff report, custom strategies
Lifetime	USD 32	Everything in Pro forever
Couples	USD 8/mo	Joint debt tracking, shared dashboard, partner sync (local network only)

6. Competitive Advantage

Feature	DebtMap	Tally	Undebt.it	Debt Payoff Planner
100% Offline	YES	NO	NO	NO
AI Financial Coaching	YES	NO	NO	NO
No Account Required	YES	NO	YES	YES
Both Strategies	YES	Avalanche	Both	Both
Single HTML File	YES	NO	NO	NO

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SOURCE CODE PAPER

Annotated technical breakdown of DebtMap.html

SOURCE CODE PAPER

DebtMap.html — Annotated Technical Breakdown

Overview

DebtMap is a single HTML file with two localStorage arrays (debts, payments), a closed-form amortisation calculator, two debt sort strategies, and one Ollama AI coaching call. The most technically interesting component is the amortisation formula that handles both interest-bearing and zero-interest debt edge cases.

Section 1: Amortisation Formula

Months-to-payoff using the closed-form amortisation equation:

```
function estimateMonths(balance, annualRate, monthlyPay) {
var r = annualRate / 100 / 12;
if(r === 0) return Math.ceil(balance / monthlyPay);
if(monthlyPay <= balance * r) return 999; // never pays off
return Math.ceil(-Math.log(1-(balance*r/monthlyPay)) / Math.log(1+r));
}
```

The r=0 guard handles interest-free debts (medical, 0% intro APR). The monthlyPay <= balance*r guard prevents Math.log(negative) NaN when minimum payment does not cover monthly interest — returns 999 as a sentinel for "infinite" payoff.

Section 2: Strategy Sort

Strategy is a single sort comparator swap:

```
var sorted = debts.slice().sort(function(a,b){
if(strategy === "avalanche") return b.rate - a.rate;
return a.balance - b.balance; // snowball
});
```

Extra monthly payment is applied only to the first debt in the sorted list (index 0). When that debt is paid off, the freed minimum payment cascades to the next — the debt payoff cascade effect both strategies rely on.

Section 3: Payment Recording & Balance Update

Payment reduces balance by principal portion (payment minus monthly interest):

```
var interest = d.balance * (d.rate / 100 / 12);
d.balance = Math.max(0, d.balance - (amount - interest));
payments.unshift({debtId, debtName, amount, date});
```

Section 4: Function Reference

Function	Role
saveDebt()	Validates and upserts debt object; initialises originalBalance for progress bar tracking
confirmPayment()	Applies payment to balance via interest subtraction; pushes to payments ledger
renderPlan()	Sorts debts by strategy; calls estimateMonths() per debt; renders ranked plan cards
estimateMonths(b,r,p)	Closed-form amortisation; r=0 guard; non-payoff sentinel 999
getAIInsight()	Builds debt summary string; calls Ollama for 3-part financial coaching response
renderSummary()	Computes total debt, total min payment, count, avg APR from debts array
renderHistory()	Renders last 50 payment records from payments array in reverse chronological order

03

HISTORY PAPER

Origin story, design decisions, and build log

HISTORY PAPER

DebtMap — Origin, Design Decisions & Build Log

Origin Story

DebtMap is Day 9 of the Mayank Creations Daily Local LLM Startup series, built July 6, 2026. Day 4 (SnapStack) covered expense tracking. Day 9 goes deeper into personal finance — specifically the highest-stakes, highest-privacy personal finance problem: debt. The local-only constraint is more important here than in any other day's build: debt data (balances, APRs, types) reveals financial vulnerability that users should never have to expose to a cloud server.

Design Philosophy

Green Palette for Financial Recovery

Green (#4ade80) is the only palette choice for a debt payoff tool — it signals growth, recovery, and money simultaneously. The dark forest green background (#0a0f0a) creates a calm, focused environment for what is often an anxiety-inducing financial task.

Both Strategies, Not Just One

Avalanche saves more money mathematically. Snowball wins psychologically. Offering both with a single toggle respects that different users have different motivational needs — and that personal finance is as much about behaviour as mathematics.

originalBalance for Progress Tracking

Debt balances decrease with each payment. To show a meaningful "% paid off" progress bar, the original balance must be stored immutably at debt creation time. This prevents the progress bar from always showing 0% after the first payment.

AI as Coach, Not Calculator

The amortisation math is computed locally by JS — no LLM is needed for numbers. The AI role is coaching: what should the user do first, how much do they save with the strategy, and what keeps them motivated. This is a deliberate use of LLM strengths (reasoning, motivation, communication) rather than tasks better handled by deterministic code.

Build Log — July 6, 2026

Time	Action	Decision
`T+0:00`	Idea selection	Personal finance — debt payoff is highest-stakes local-only use case in the series
`T+0:15`	Data model	debts[] + payments[] as two separate arrays; originalBalance immutable at creation
`T+0:35`	Amortisation formula	Closed-form log equation; added r=0 guard for 0% APR; 999 sentinel for non-payoff

T+0:55	Strategy toggle	Avalanche vs Snowball as a single sort comparator swap; extra payment to rank-1 debt only
T+1:15	Progress bar	(1 - balance/originalBalance) * 100; requires originalBalance stored at creation
T+1:30	AI coaching prompt	3-part: biggest action + interest saved + motivation; sends debt summary string only (no personal in
T+1:50	Payment modal	Pre-fills minPay amount; updates balance via interest-subtraction formula
T+2:05	GitHub + LinkedIn	Links included per series standing instruction
T+2:20	Documentation	Whitepaper, Source Code Paper, History Paper compiled into single PDF

Key Decisions Rejected

Rejected	Why Rejected
Bank account linking	Directly contradicts local-only privacy mission
Compound interest simulation (full amortisation table)	Month-by-month simulation accurate but overkill for payoff date estimates; closed-form formula
Debt-to-income ratio calculator	Requires income input — adds scope; deferred to Pro tier
Notification reminders for payment due dates	Web Notifications API available but payment due dates vary; deferred to v2
Chart/graph of balance over time	Canvas debt payoff curve adds visual value but increases code complexity; deferred to Pro

What Day 10 Could Build On

- Balance-over-time chart using Canvas 2D
- Debt-to-income ratio and credit utilisation tracker
- Payment due date reminders via Web Notifications API
- PDF payoff report export (per debt amortisation table)
- Net worth tracker combining assets and liabilities
- Couples/shared debt mode via local network sync

DebtMap — Day 09 of the Daily Local LLM Startup Series

Built by Mayank Swaraj | Mayank Creations | July 6, 2026

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